

SUNSPOKE · DESIGN DECISIONS REGISTER

SunSpoke design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from the decision records, the review note and the design for construction

SunSpoke design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction	Accept all changes P1 to P13; accept some; ask for others	Accept all	Every made part and joint in the build plan	SSP-DDR-003, Table 1
2	Pack swap at the cradle: two movements (open the latch, fold the gate) where the concept showed one lever	(a) gate and draw latch as modelled; (b) design a single lever that also folds the gate	(a) for the prototype; review after the first swap trials	Gate, hinge and latch (build plan 3.6, step 12)	SSP-DDR-003, A1
3	Panel stand material	(a) timber, as modelled; (b) steel angle, bolted	(a)	Stand (build plan 3.8, steps 13 to 15)	SSP-DDR-003, A2; review note 2026-09-26, proposal 4
4	Whether light filing of the donor fork slots (about 0.24 mm per side) is acceptable with torque arms fitted (R4)	Allow filing with torque arms; never file (choose donors whose slots take 10 mm flats)	None yet; needs the donor survey. Until then the build plan asks for a donor that needs no filing	Choice of donor; torque arms	SSP-DDR-001, item 15
5	First co-design partner and region	Chosen per area later	None (portfolio rule)	Donor survey, load cases, field trials (TRL 4 and later)	SSP-DDR-001, item 14
6	Appearance model: cranks turned vertical so the rider's feet sit on the pedals	Accept; keep cranks as in <code>model.py</code>	Accept	Renders only	Review note 2026-09-26, proposal 1
7	Appearance model: saddle top raised about 12 mm to meet the mannequin	Accept; lower the mannequin	Accept (within normal saddle adjustment)	Renders only	Review note 2026-09-26, proposal 2
8	Brake sensors moved under the bar beside the levers	Accept; keep on the grip	Accept; <code>model.py</code> now matches (SSP-DDR-003, P10)	Display and brake sensors (build plan 3.9.3)	Review note 2026-09-26, proposal 3
9	Hero render shows the rider seated while the pack charges	Keep; pose a standing owner by the parked bike	Keep; revisit if it reads oddly	Renders only	Review note 2026-09-26, proposal 5
10	Wet braking: whether R11 should add a wet target, and whether better blocks or a rim brake upgrade belong in the kit	Add a wet target and a brake upgrade; keep R11 dry with a no-wet-riding rule for trials	None yet	Brakes (build plan section 1 and S7)	SSP-PRC-001, open questions; SSP-CAL-001 section 10

#	Decision needed	Options	Recommendation	Affects in the build	Source
11	Motor data for R3: a named 250 W geared hub with winding resistance, thermal capacity and gear temperature limit	Name a motor and rerun R3	None yet	Motor wheel (line 1)	SSP-PRC-001, open questions

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The motor's axle has 10 mm flats, an M12 thread and enough length for three threads past the nut over the dropout, torque arm and washer	The torque arms and dropouts are sized to it	SSP-DDR-003, P7
2	The donor's dropout slots take the 10 mm flats, its down tube is round and 28 to 32 mm, and the step from dropout face to blade outside is about 8 mm	The saddles, torque arm slot and joggle are sized to these	SSP-DDR-003, P1 and P7
3	The donor's bottom bracket is cup-and-cone with a left lockring	The pedal-assist bracket fits under it	SSP-DDR-003, P9
4	The draw latch: adjustable hook, safety catch, 500 N working load or more, over centre with 50 N or less at its lever	It sets the V1 preload of 330 N	SSP-DDR-003, P4; SSP-CAL-001 section 8
5	The band clamp torque that gives 1.5 kN of band tension	The cradle retention margins assume it	SSP-CAL-001 section 8
6	The SwapCell receptacle's flange and hole pattern	The end stop's four holes are drawn for a 37 x 24 mm pattern	SSP-DDR-003, P3
7	The host adapter's base inserts (two, 40 mm apart) and the controller case size	The tray holes and the band clamps are sized to them	SSP-DDR-003, P6 and P8
8	The panel's mounting hole positions on its frame	The stand's rail holes are drilled to match	SSP-DDR-003, P12
9	The pack moves from legacy discharge (state 5) to heartbeat discharge (mode 2) without opening its output	The bike relies on it if the adapter is slow to start; raised with SwapCell	SSP-DDR-002, item 18
10	A low-cost boost charger with input-voltage tracking gives near-MPPT yield from one 18 V panel	R8 assumes 92 %	SSP-PRC-001, open questions

■ Value engineering

Value-engineering target: USD 250 for the bike kit (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 196 for the bike kit (USD 54 under the target). The solar set is costed separately at USD 90, and the SwapCell pack (USD 414) is priced once in the SwapCell project. Main cost drivers and savings worth trying:

- The largest lines are the motor wheel (USD 72), the receiver cradle (USD 33), the controller (USD 25), the harness (USD 15), the host adapter (USD 14) and the handlebar display and brake sensors (USD 14).
- Making the design constructable added USD 8 to the bike kit (cradle USD 28 to 33, hardware USD 8 to 11) and USD 4 to the solar set (stand USD 10 to 14).
- Savings worth trying: buy the receptacle, draw latch and hinge in batches with other SwapCell vehicle receivers; fold the tray from 2 mm aluminium if the retention and end stop margins still hold; a shared panel and stand at a village hub removes the USD 90 solar set for riders who swap packs there.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	48 V on the SwapCell pack; budget kept at USD 250 with R12 covering the bike kit and the solar set costed separately; geared front hub; pack on the down tube; local lacing; 20 km/h default cut-off; pedal assist and walk assist, no throttle; charge on the bike; SwapCell interface v0.3 items W, C and V; pack priced once in SwapCell	Amish: go with recommendation	SSP-DDR-001, items 1 to 13
2026-09-25	Handlebar power switch in the INTERLOCK loop; motor thermistor with a controller derate instead of a cutout; raise the legacy-to-heartbeat transition with SwapCell	Amish: "i accept all your recommendations, go with them across all repos."	SSP-DDR-002, items 16 to 18
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	SSP-DDR-003 (Draft, open for review: open decision 1)
2026-10-01	Treat <code>budget_usd</code> as a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	STANDARDS section 18; this register's Value engineering section